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Data Analyst Job Market Analysis

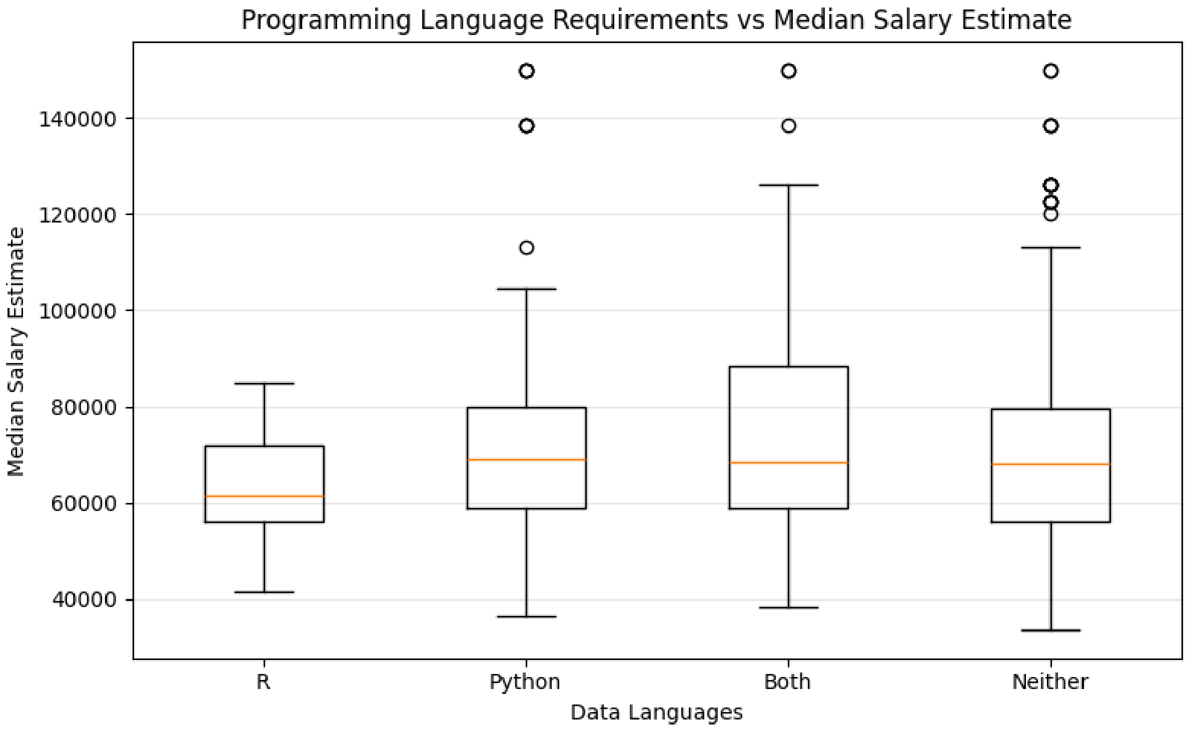
Introduction

This project examines a sample of Data Analyst job postings to explore how experience requirements and programming language requirements relate to salary estimates. The analysis focuses on median salary estimates, minimum years of experience, and whether postings require R, Python, both languages, or neither. The goal is to identify visible patterns in the job market data while recognizing that the dataset is only a sample and may not represent the full Data Analyst job market.

Years of Experience Requirement Analysis (Scatterplot interpretation):

The scatter plot shows **Minimum Years Experience** with **Median Salary Estimate**. Overall, there is not a perfectly clear linear relationship between experience and salary in this sample. Salaries vary widely at similar experience levels, especially between 1 and 5 years of experience. This suggests that salary may also be influenced by other factors, such as company, location, industry, job title, and technical requirements.

A limitation is that **91 postings had missing or non-numeric experience values**, so they could not be included in this plot. Also, because this is a relatively small sample of job postings, the pattern should not be treated as a definitive market trend.



Programming Language Analysis (Box plot interpretation):

The box plot compares **Median Salary Estimate** across the four **Data Languages** requirement groups. The Python, Both, and Neither categories have similar median salaries, while the R-only group appears slightly lower. The R-only category has the smallest number of postings only 15, so its salary distribution is less reliable than the larger groups. The Neither category has the largest sample size and shows a wide salary range, suggesting that many Data Analyst postings may emphasize tools other than R or Python, such as SQL, Excel, Tableau, or domain-specific platforms.

The differences between groups should not be treated as definitive because the sample size is limited, especially for the R category, and the analysis does not control for other salary-related factors.

Personal Reflection:

This week's process for the assignment was quite daunting and seemed long at first sight. However, it became a doable process with the help of the pre-work exercises and the sets of instructions provided. I'm really impressed with having access and using the API key as I trialled using Gemini and Perplexity sonar and there were only 5 differences from the 400 sets of the job posting. Gemini gave a more accurate response for data language requirements in the assignment above. This just opens up many possibilities in data analysis and research while making the process less time-consuming.

I learnt a lot about making the prompt give exactly the information I needed. There was some trial and error in fine tuning the prompt (in this case to give both R and Python, if it was stated in the posting) and then verifying with the small sample whether it was accurate but I got there in the end. I'm gradually becoming more familiar with writing effective prompt and will only improve with continued practice.

Before taking the course, I was very reserved with using AI tools due to concerns about privacy, data security and accuracy of the data generated. However, each week has taught me ways to minimize the risks and how to validate and verify the data before using it. I see the positive impact it has on research analysis and many other applications especially with my line of work in the health sector.